CIE AS PHYSICS (9702) • MARK SCHEME & MODEL ANSWERS

1.4 Scalars & Vectors

Syllabus Section: 1. Physical Quantities & Units **Paper Type:** Multiple Choice (Paper 1) **Total:** 16 Questions (16 Marks)

Question 1 Easy [1 mark]

Which pair contains one vector and one scalar quantity?

A	Displacement	Acceleration
B	Force	Kinetic energy
C	Momentum	Velocity
D	Power	Speed

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **B** because:*

- Force is a vector quantity as it has magnitude and direction
- Kinetic energy is a scalar quantity as it has only magnitude

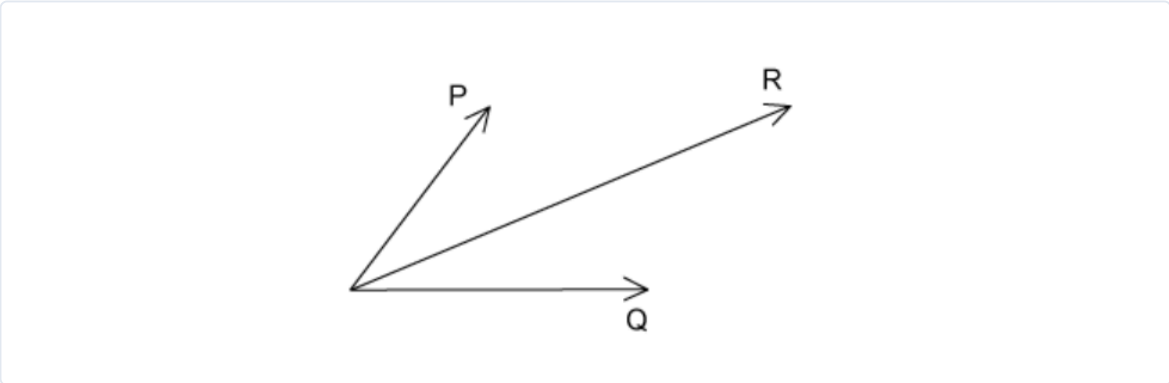
A is incorrect as displacement and acceleration are both vector quantities

C is incorrect as momentum and velocity are both vector quantities

D is incorrect as power and speed are both scalar quantities

Question 2 Easy [1 mark]

Two physical quantities **P** and **Q** are added together. The sum of **P** and **Q** is **R**, as shown in the diagram.



Which quantity could be represented by **P** and by **Q**?

A Kinetic energy

B Power

C Speed

D Velocity

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- The diagram represents a quantity which has a magnitude and direction, so it must be a vector
- Velocity is the only vector quantity listed

A, B and **C** are incorrect as kinetic energy, power and speed are all scalar quantities as they all have a magnitude but no direction

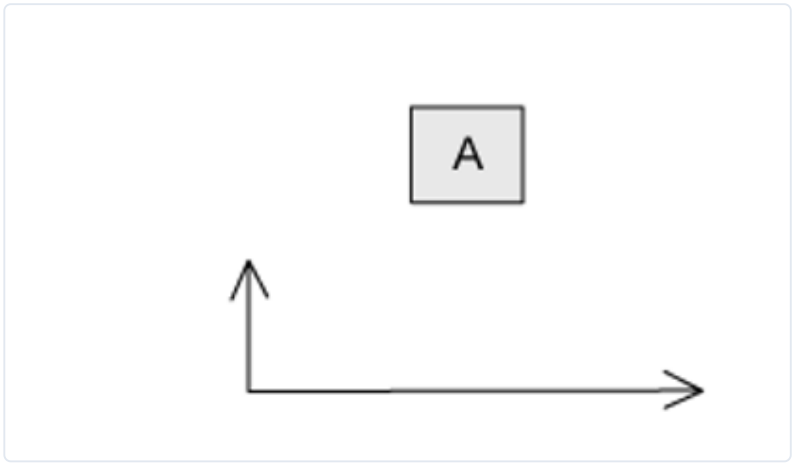
Question 3 Easy [1 mark]

The arrow represents the vector **R**.

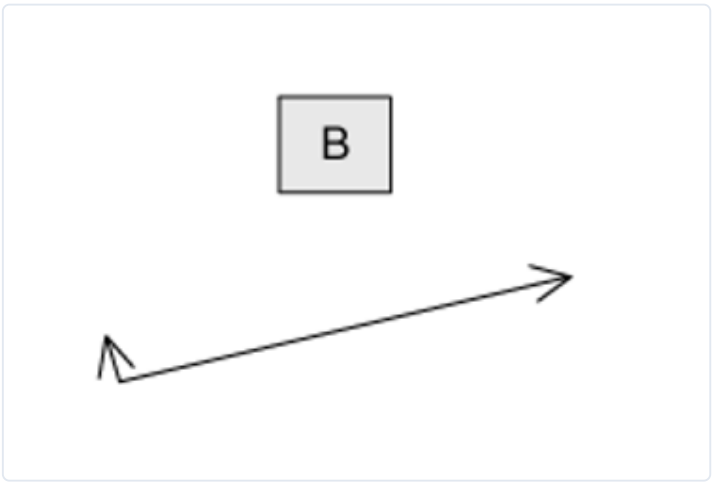


Which diagram does **not** represent **R** as two perpendicular components?

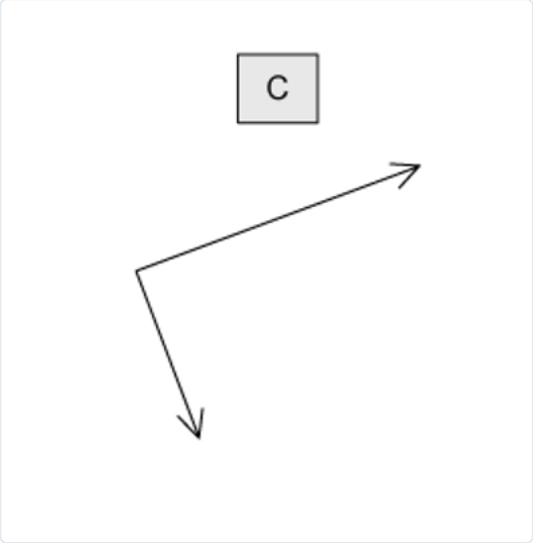
A



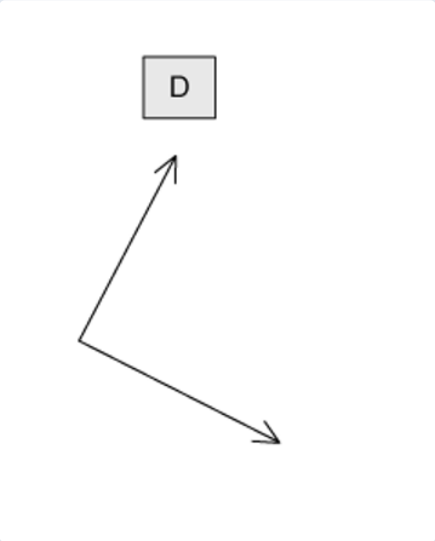
B



C



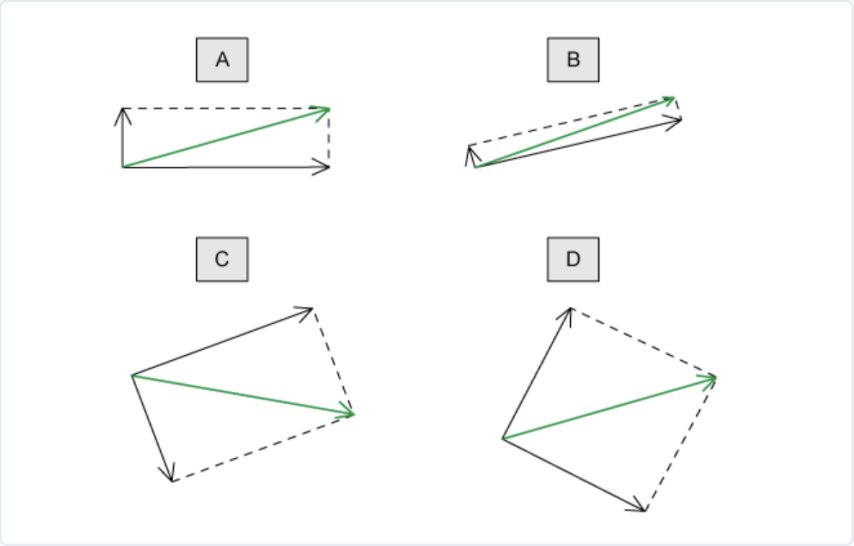
D



Mark Scheme & Explanation:

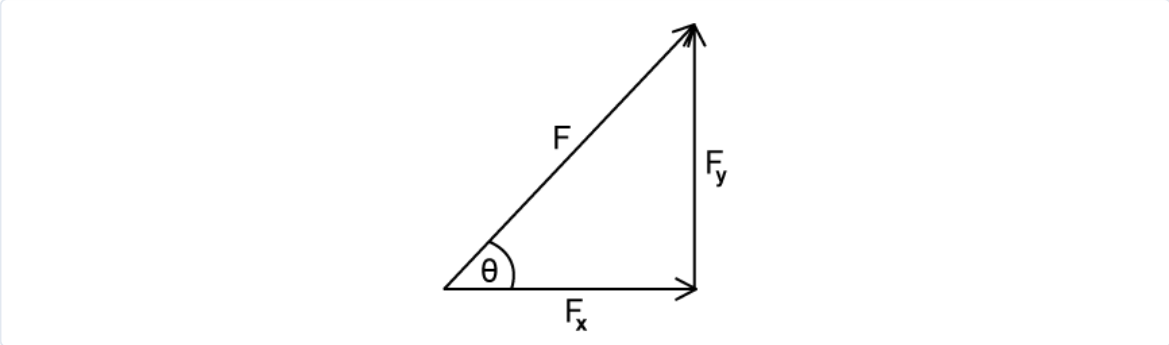
*The correct answer is **C** because:*

- The vector sum of the two perpendicular components should provide a resultant vector equal to the vector **R**
- If we complete the diagrams, we can see that option C is the odd one out as it produces a resultant vector in a downward direction, therefore it cannot be equal to vector **R**


A
B
C
D

Question 4 Easy [1 mark]

The diagram shows a resultant force *F* at an angle of *θ* to the horizontal. The horizontal component F_x and vertical component F_y are labelled on the diagram.



What is the horizontal component of *F*?

A $F \sin \theta$

B $F \cos \theta$

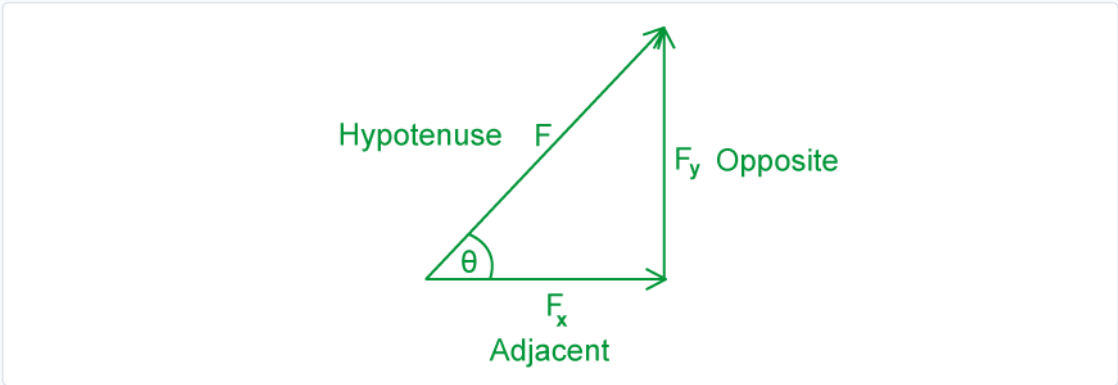
C $F \tan \theta$

D $F_y \cos \theta$

Mark Scheme & Explanation:

*The correct answer is **B because:***

- The hypotenuse, adjacent and opposite sides to the angle can be labelled as follows



- The horizontal component is
  - $\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$

- $\text{adjacent} = \text{hypotenuse} \times \cos\ \theta$
- $F_x = F \cos\ \theta$

- This is option **B**

You can use the mnemonic SOH CAH TOA as a checklist to see which trigonometric ratio you need to use. In this question, we know the hypotenuse (the resultant force), and we need to find the adjacent (the *x* component), so we use CAH, which tells us that cos *θ* is equal to the adjacent over the hypotenuse.

Question 5 Easy [1 mark]

Which one of the following statements is **true**?

A A scalar quantity has both magnitude and direction

B A scalar quantity has only direction

C A vector quantity has only magnitude

D A vector quantity has both magnitude and direction

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- A vector quantity has both magnitude and direction

Option **A** and **B** is incorrect because

- A scalar quantity only has magnitude

Option **C** is incorrect because

- A vector quantity has direction as well as magnitude

Question 6 Medium [1 mark]

The speed of an aeroplane in still air is 200 km h^{-1}. The wind blows from the west at a speed of 85.0 km h^{-1}.

In which direction must the pilot steer the aeroplane in order to fly due north?

A 23.0° east of north

B 23.0° west of north

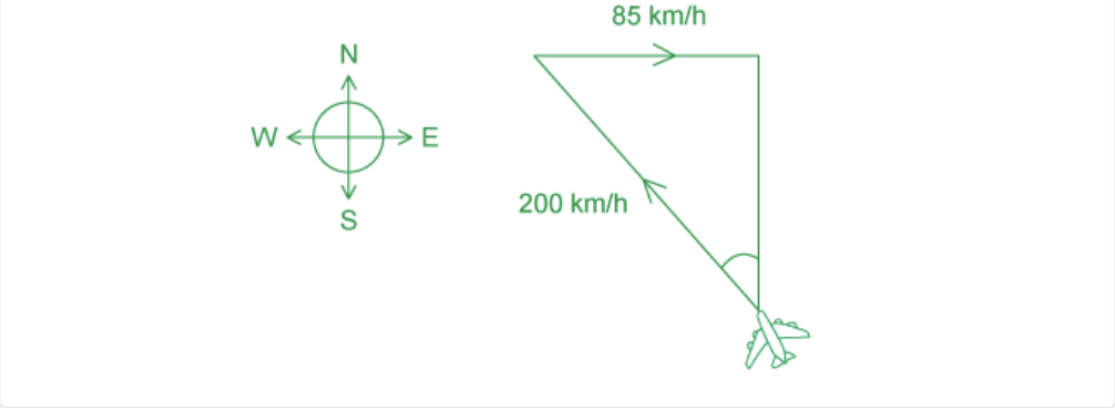
C 25.2° east of north

D 25.2° west of north

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- When we construct a vector triangle, we can see the direction will be west of north
- We can calculate the angle from
  - $\sin^{-1} \frac{85}{200} = 25.15°$



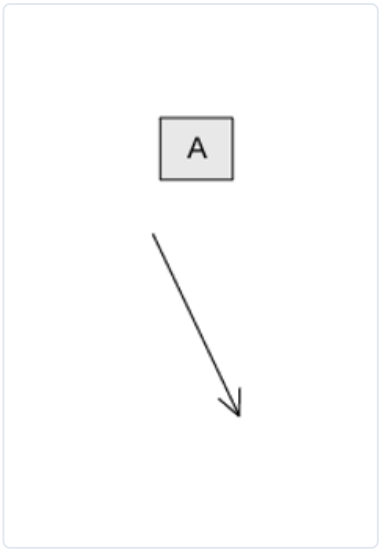
Question 7 Medium [1 mark]

Vectors **P** and **Q** are drawn to scale.

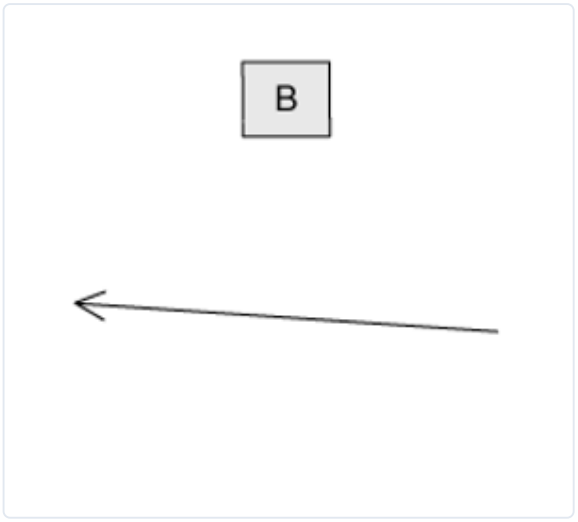


Which diagram represents the vector (**P** – **Q**)?

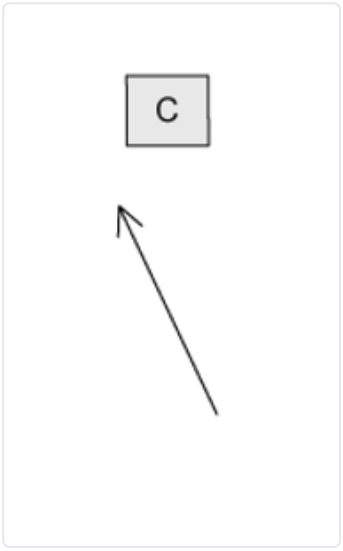
A



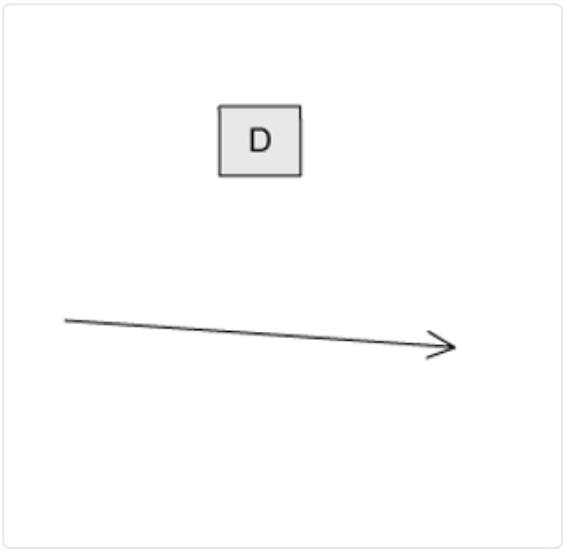
B



C



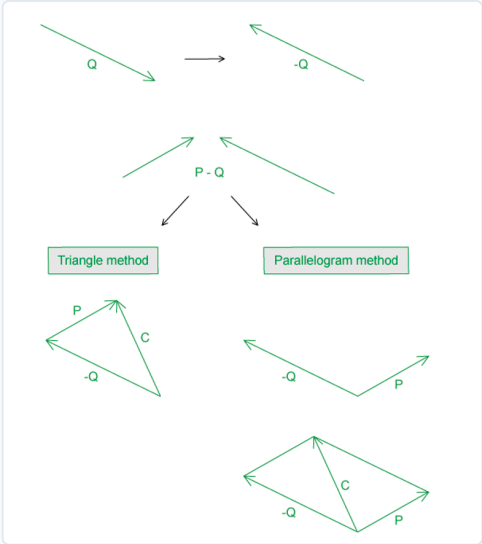
D



Mark Scheme & Explanation:

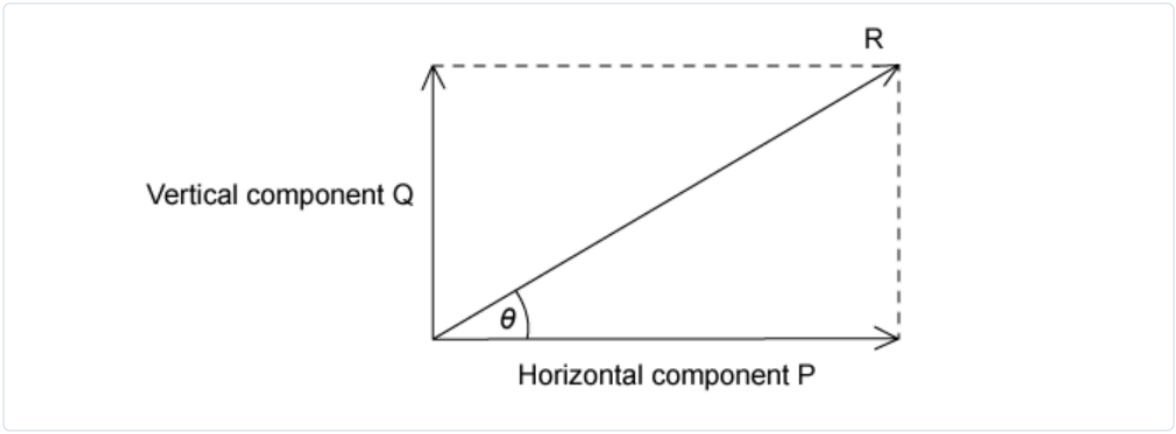
*The correct answer is **C** because:*

- First, it's important to notice that the vector **Q** is negative so its direction must be reversed
- Then using either the triangle or parallelogram method it's possible to determine the direction of the resulting vector, as shown in the diagram below


Q
-Q
P - Q
Triangle method
Parallelogram method
P
C
-Q
-Q
P
C
-Q
P

Question 8 Medium [1 mark]

A vector has magnitude **R** and perpendicular components **P** and **Q**, as shown in the diagram.



Which row correctly describes the perpendicular components?

	Vertical component	Horizontal component
A	Q	$\sin\theta$
B	$R\cos\theta$	P
C	$R\cos\theta$	$R\sin\theta$
D	$R\sin\theta$	$R\cos\theta$

A

B

C

D

Mark Scheme & Explanation:

*The correct answer is **D** because:*

- Vertical component
  - $\sin\theta = \frac{opposite}{hypotenuse} = \frac{Q}{R}$
  - $Q = R\sin\theta$
- Horizontal component
  - $\cos\theta = \frac{adjacent}{hypotenuse} = \frac{P}{R}$

- $P = R \cos \theta$

Question 9 Medium [1 mark]

A resultant force of 20 N is 30° to the horizontal.

What is the horizontal and vertical component of the resultant force?

	Horizontal Component / N	**Vertical Component / N**
A	17	10
B	10	17
C	11.5	10
D	17	11.5

A

B

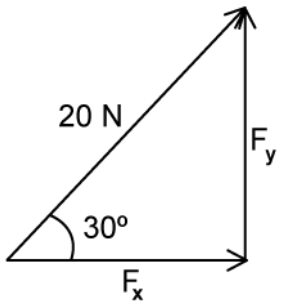
C

D

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The vector diagram will look as follows

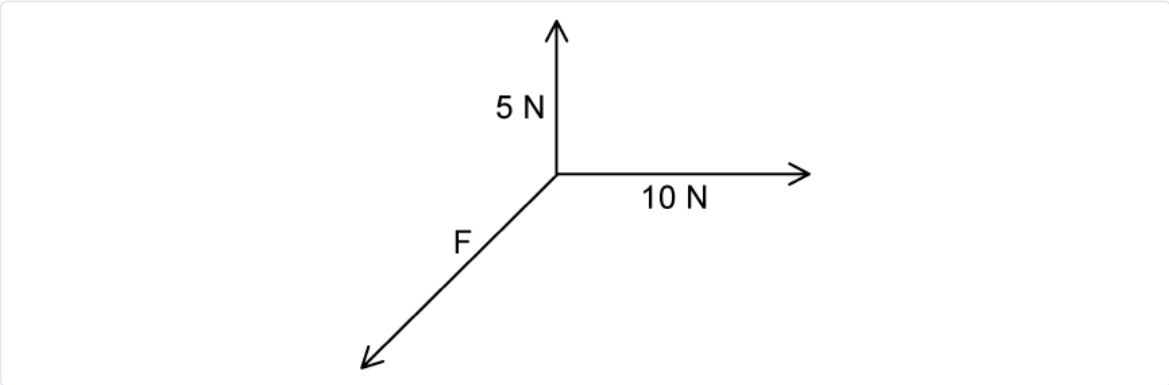


- The horizontal component is
  - $F_x = F \cos \theta$
  - $F_x = 20 \times \cos (30) = 17 \text{ N} (2 \text{ s.f})$

- The vertical component is
  - $F_y = F \quad \sin \quad \theta$
  - $F_y = 20 \times \sin \quad (30) = 10 \quad \mathrm{N}$
- This is option **A**

Question 10 Medium [1 mark]

Fig 1.1 shows three coplanar forces which are in equilibrium.



What is the value of force *F*?

A 5 N

B 8.7 N

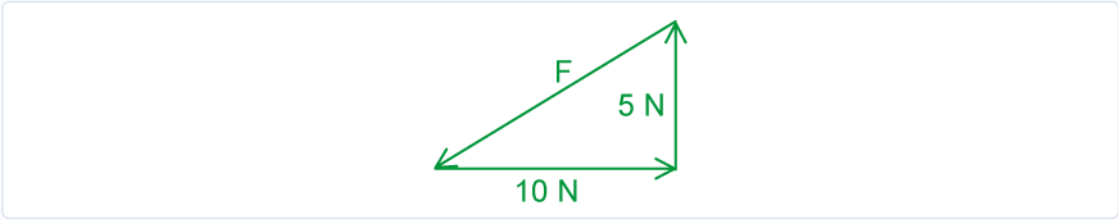
C 11 N

D 15 N

Mark Scheme & Explanation:

*The correct answer is **C** because:*

- When three forces are in equilibrium they form a closed triangle
- By sliding the vectors past each other so they are tip to tail, the vector diagram will look as follows



- As the 5 N and 10 N forces are perpendicular to each other, pythagoras theorem can be used to calculate the hypotenuse, *F*
  - $a^2 = b^2 + c^2$
  - $a = \sqrt{(b^2 + c^2)}$
- Substituting in 5 N and 10 N

- $F = \sqrt{(5^2 + 10^2)} = 11\text{N}$ (2 s.f)

- This is option **C**

Question 11 Medium [1 mark]

A ship's speed in still water is 20 km h^{-1}. A current coming from the east pushes the ship at a constant speed of 5 km h^{-1}. At what bearing should the captain direct the ship so that it travels perfectly due north?

A 014.5°

B 075.5°

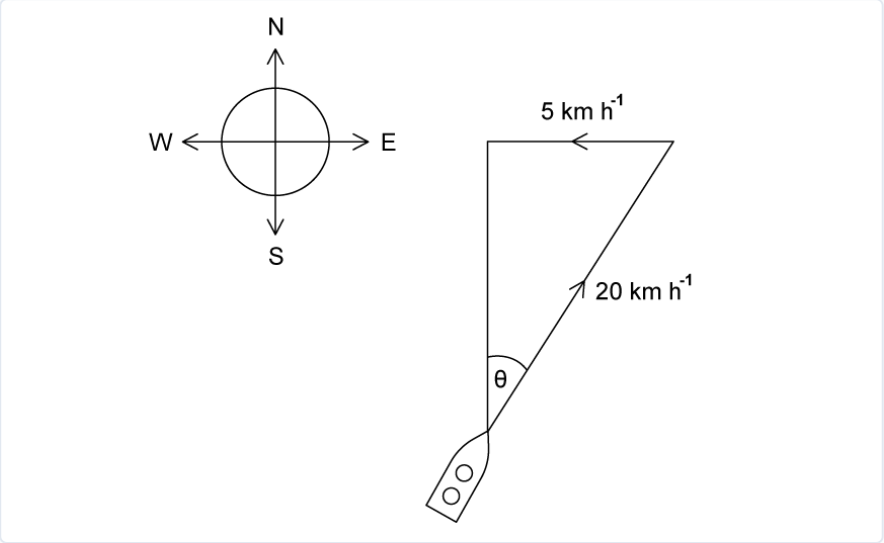
C 284.5°

D 345.5°

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- Construct a vector diagram and label the triangle



- Consider SOH CAH TOA:
  - We know the opposite and hypotenuse and want to find the angle
  - So, we must use $sin\ \theta = \frac{Opposite}{Hypotenuse}$
- The correct angle is given by $sin\ \theta = \frac{5}{20}$
- Rearrange to find the value of sin θ:
  - $sin^{-1}\frac{5}{20} = 14.5°$
- Convert into a bearing:

- 14.5° = 014.5°

B is incorrect because cosine has been used instead of sine.

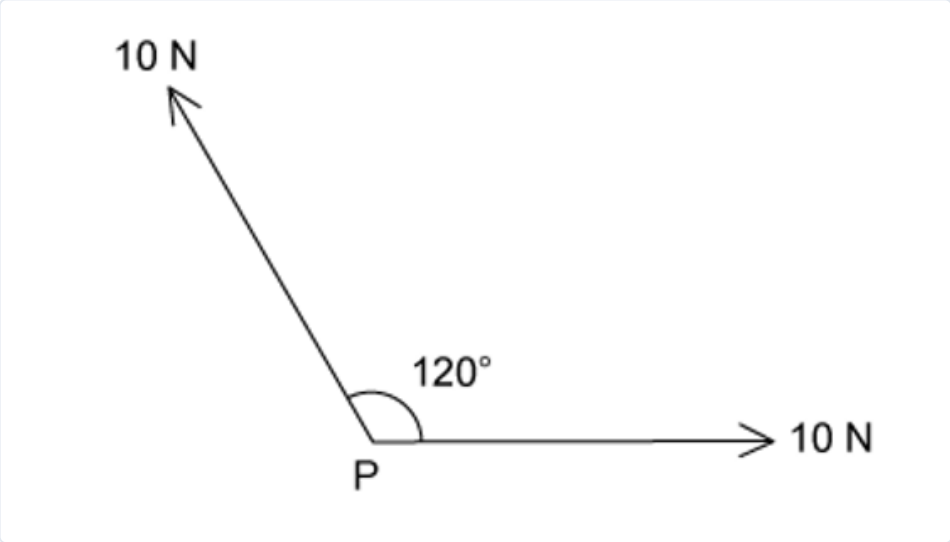
C is incorrect because, although the angle away from north is correct, this bearing does not oppose the direction of the current, so the ship would not head northwards.

D is incorrect, both because cosine has been used instead of sine, and because the bearing direction does not oppose the direction of the current.

Don't be fooled by the units of km h^{-1}. You don't need to convert them into standard units for this question because you are using trigonometry to find a ratio. Label the sides of your triangle clearly with opposite, adjacent and hypotenuse, so you choose the correct ratio.

Question 12 Hard [1 mark]

Two forces, each of 10 N, act at a point *P* as shown in the diagram. The angle between the force vectors is 120°



What is the magnitude of the resultant force?

A 5 N

B 10 N

C 17 N

D 20 N

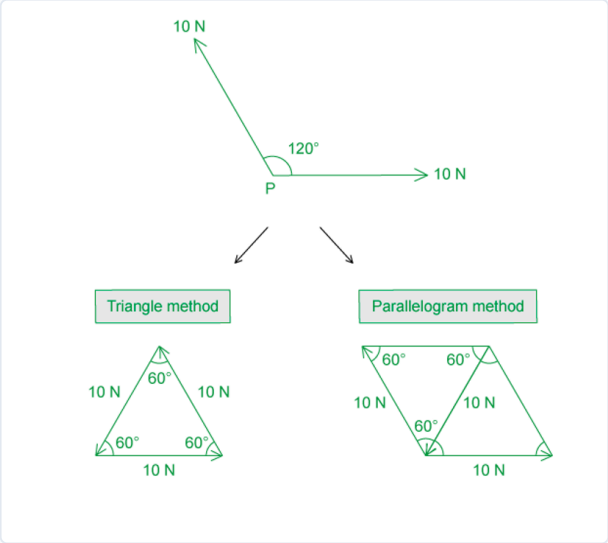
Mark Scheme & Explanation:

*The correct answer is **B** because:*

Method 1

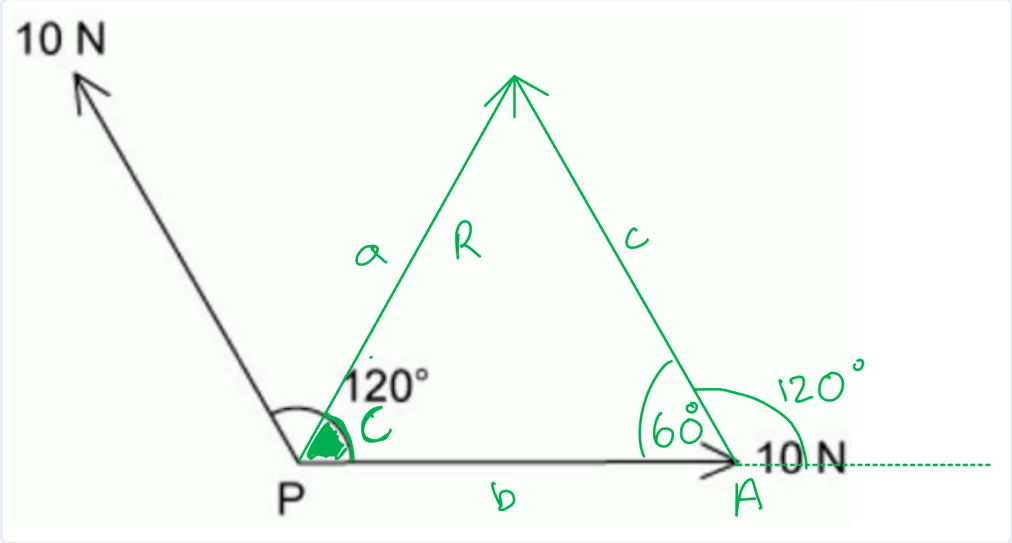
- Using either the triangle or parallelogram method it's possible to determine the direction of the resulting vector
    - In both methods, you must slide the vectors, parallel to their current direction, so that they are arranged tip to tail
    - The resultant vector is the one now joining from tail to tip on the other side
    - The 120° angle will be on the outside of the triangle or parallelogram
        - So, the interior angle = 180 −120 = 60°
- As shown in the diagram below, you can see that:

- The vector triangle formed is an equilateral triangle
- Where each angle is equal to 60° and the magnitude of each vector is 10 N
  - The resultant vector is the same length as the 10 N vectors, as the diagram is drawn to scale



Method 2

- Consider the cosine rule to find the length of the resultant side:
  - $a^2 = b^2 + c^2 - 2bc\ \cos A$
  - Label the angles and sides of the triangle



- Substitute in the known sides and angle
  - $a^2 = 10^2 + 10^2 - 2 \times 10 \times 10 \times \cos 60$
- Calculate the length of the resultant side *a*:
  - *a* = 10 N

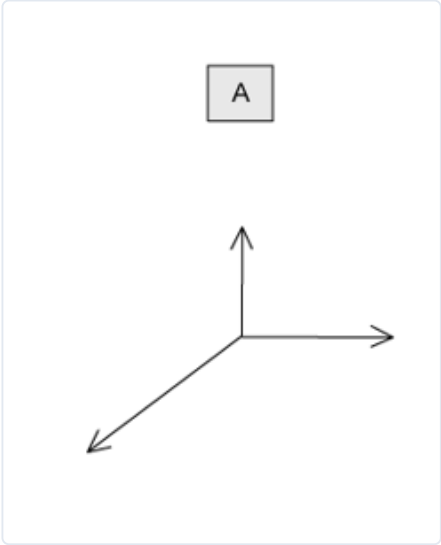
- Consider the sine rule to find the angle of the resultant side to the horizontal:
  - $\frac{a}{\sin A} = \frac{c}{\sin C}$
- Substitute in the known sides and angle:
  - $\frac{10}{\sin 60} = \frac{10}{\sin C}$
- Rearrange the equation and calculate the angle of the resultant side to the horizontal:
  - $10 \times \sin C = 10 \times \sin 60$
  - $\sin C = \sin 60$
  - So, $C = 60°$

Question 13 Hard [1 mark]

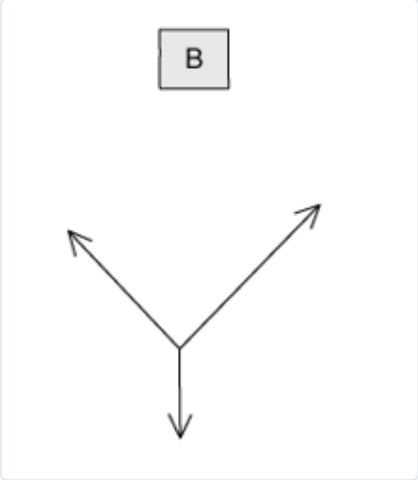
The diagrams represent systems of coplanar forces acting at a point. The lengths of the force vectors represent the magnitudes of the forces.

Which system of forces is in equilibrium?

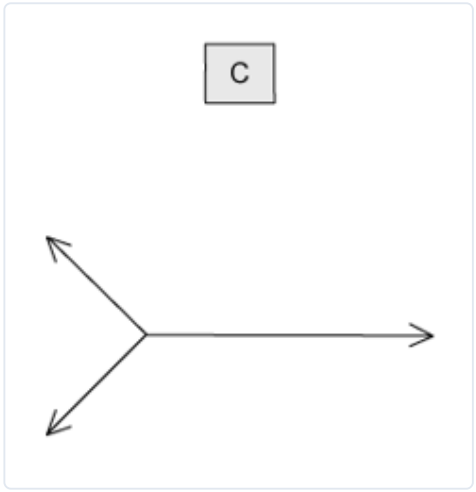
A



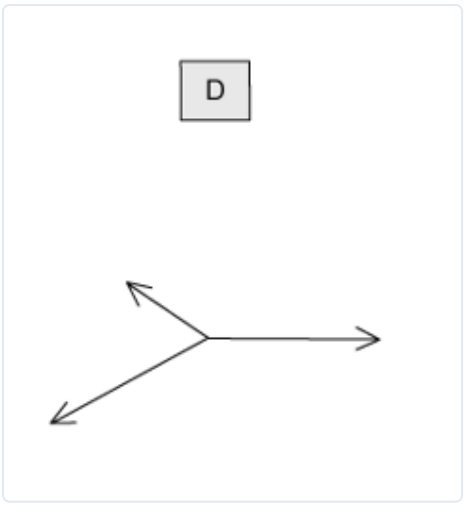
B



C



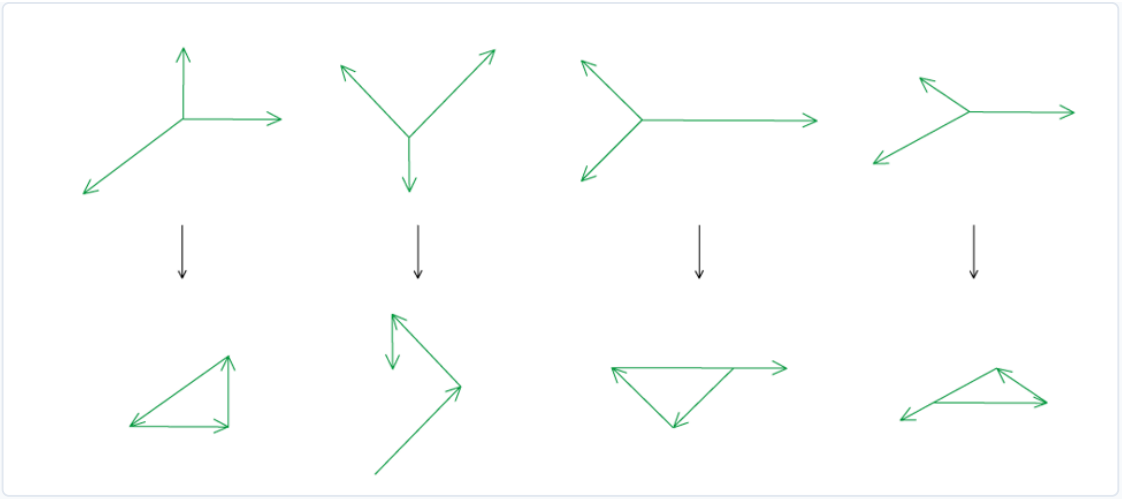
D



Mark Scheme & Explanation:

*The correct answer is **A** because:*

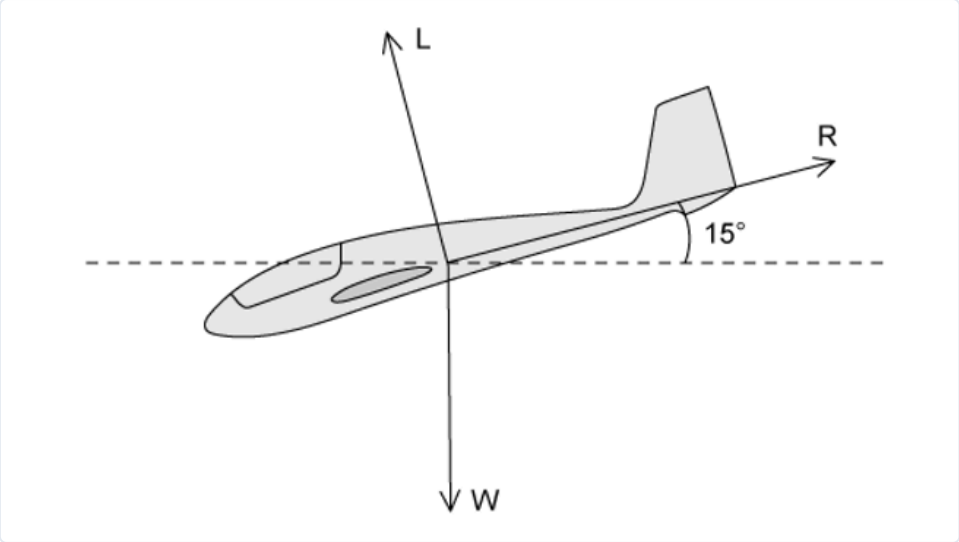
- Equilibrium means the forces are balanced. For 3 forces in equilibrium, the force vectors always form a closed triangle
- Constructing vector triangles (to scale, therefore, it is recommended you use a ruler), we can see the only balanced system is A
- This is because the sizes and directions of the forces align and form a closed triangle, satisfying the condition that the net force must be zero



B, C & D are incorrect because none of these systems of coplanar forces form a closed triangle meaning the net force is not zero

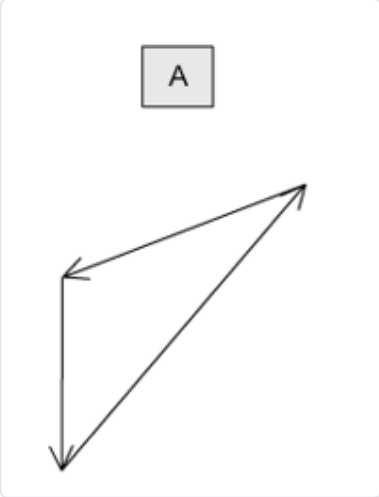
Question 14 Hard [1 mark]

A glider is descending at a constant speed at an angle of 15° to the horizontal. The diagram shows the directions of the lift **L**, air resistance **R** and weight **W** acting on the glider.

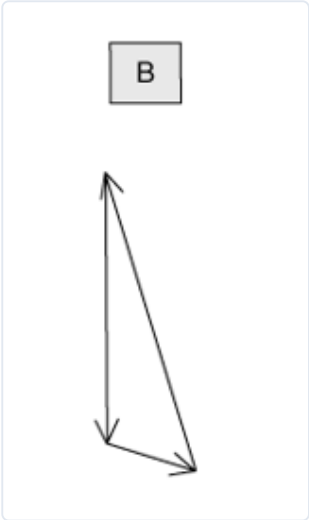


Which vector triangle could represent the forces acting on the glider?

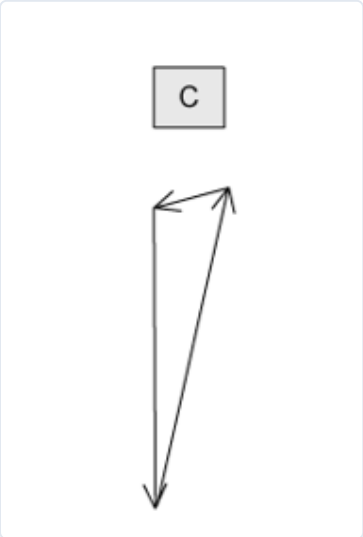
A



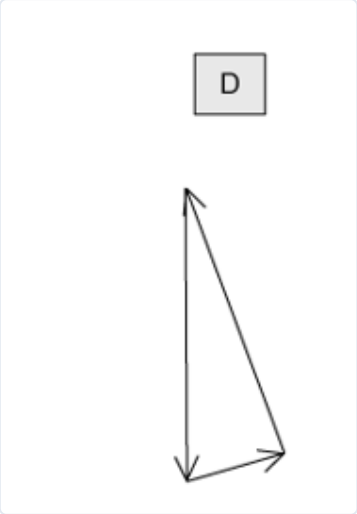
B



C



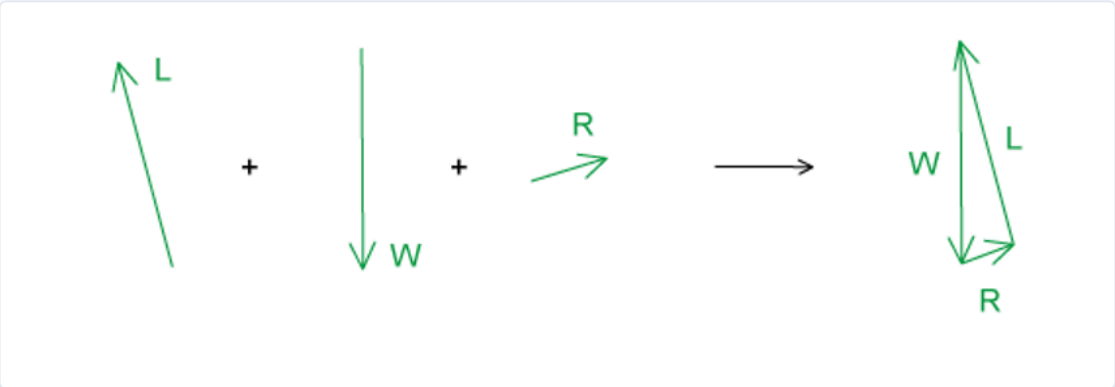
D



Mark Scheme & Explanation:

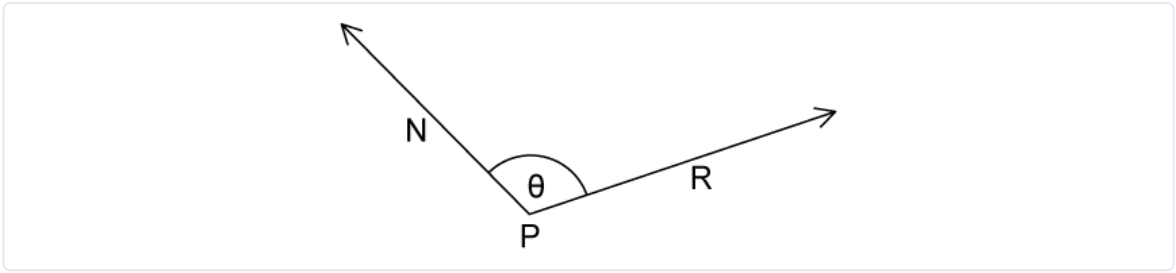
*The correct answer is **D** because:*

- When you examine the individual vectors in the scenario you can immediately whittle the selection down to either B or D as both correctly represent vectors **L** and **W**
- On further inspection, you can see vector **R** is inclined 15° to the horizontal in a north-west direction so the correct answer is D



Question 15 Hard [1 mark]

Figure 1.1 shows two coplanar forces, *R* and *N*, which act at point *P*. The angle between the forces is θ, where $\theta > 90$.



What is the magnitude of the resultant force *F* equal to?

A $\sqrt{R^2 + N^2 - F^2}$

B $\sqrt{R^2 + N^2 - 2RN \cos (180 - \theta)}$

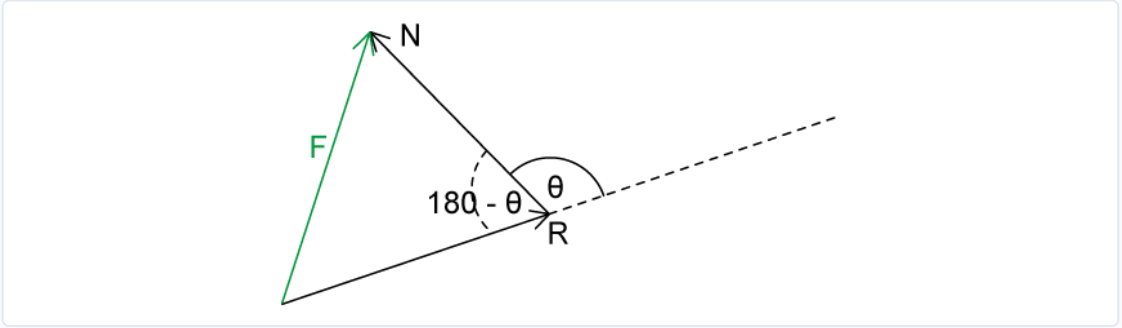
C $\sqrt{R^2 + N^2 - 2RN \cos (\theta)}$

D $R^2 + N^2 - 2RN \sin(F)$

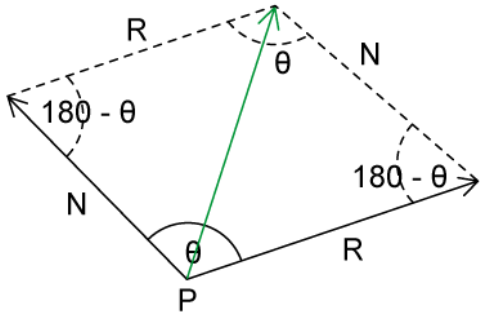
Mark Scheme & Explanation:

*The correct answer is **B** because:*

- The direction of the resultant force *F* can be found by using either the triangle or the parallelogram method
  - This is because $\theta > 90$
- For the triangle method
  - Slide the vectors past each other so they are tip to tail
  - The angle θ will now be on the outside of the triangle
  - The resultant vector will be tail to tail and tip to tip



- For the parallelogram method
  - Draw lines from the tips of the vectors so they are parallel and equal to the opposite vector
  - The resultant vector will come from point *P* to where the lines meet



- The cosine rule can now be used to find the magnitude of the resultant force *F*
  - $a^2 = b^2 + c^2 - 2bc \quad \cos(A)$
  - $F^2 = R^2 + N^2 - 2RN \quad \cos(180 - \theta)$
  - $F = \sqrt{R^2 + N^2 - 2RN \quad \cos(180 - \theta)}$
- This is option **B**

Question 16 Hard [1 mark]

A boat is being pulled along by two equal forces of 700 kN, both at angle θ to the direction of motion. The boat experiences a resistive force of 40 kN. The net force on the boat is 1270 kN.

What is the value of θ?

A 21°

B 25°

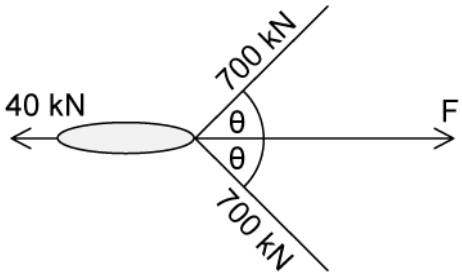
C 57°

D 61°

Mark Scheme & Explanation:

*The correct answer is **A** because:*

- The known quantities are
  - Net force, F_{net} = 1270 kN
  - Resistive force, F_{R} = 40 kN
  - Single force pulling the boat, F_{pull} = 700 kN
- The free-body diagram of the boat will look as follows



- The resultant force *F* from the forces pulling the boat is equal to
  - $F_{\text{net}} = F - F_{\text{R}}$
  - $F = F_{\text{net}} + F_{\text{R}} = 1270 + 40$
  - $F = 1310 \text{ kN}$
- By resolving the forces that are pulling the boat
  - $2 \times 700 \cos \theta = 1310$

- $\cos\ \theta\ =\ \frac{1310}{2\times 700}\ =\ 0.9357$

- The angle θ is
  - $\theta\ =\ \cos^{-1}\ 0.9357$
  - $\theta\ =\ 20.656\ \approx\ 21°$
- This is option **A**